Practical3

August 9, 2026

1 Practical No. 3 — Sets & Dictionary

Software Lab I | III Semester (CSE) | Session 2026–2027

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5.1 Question 1

[1]:
```
import random

s = set()

while len(s) < 15:
    s.add(random.randint(20, 80))

print("Original set:", s)

even = 0
greater = 0

for i in s:
    if i % 2 == 0:
        even += 1
    if i > 50:
        greater += 1

print("Even numbers:", even)
print("Numbers greater than 50:", greater)
print("Maximum:", max(s))
print("Minimum:", min(s))
```

```
s = {i for i in s if i >= 35}

print("Modified set:", s)
print("Sum:", sum(s))
print("Average:", sum(s) / len(s))
```

```
Original set: {32, 66, 26, 67, 37, 70, 72, 42, 43, 29, 21, 54, 58, 61, 30}
Even numbers: 9
Numbers greater than 50: 7
Maximum: 72
Minimum: 21
Modified set: {66, 67, 37, 70, 72, 42, 43, 54, 58, 61}
Sum: 570
Average: 57.0
```

5.2 Question 2

[2]:
```
n = 500

F = 49 * n // 100
H = 53 * n // 100
B = 62 * n // 100

FH = 27 * n // 100
HB = 29 * n // 100
FB = 28 * n // 100

none = 5 * n // 100
atleast_one = n - none

all_three = atleast_one - F - H - B + FH + HB + FB

only_F = F - FH - FB + all_three
only_H = H - FH - HB + all_three
only_B = B - FB - HB + all_three

only_one = only_F + only_H + only_B
atleast_two = FH + HB + FB - 2 * all_three

print("i.", all_three)
print("ii.", only_F, ":", only_H)
print("iii.", only_one)
print("iv.", atleast_two)
```

```
i. 75
ii. 45 : 60
iii. 205
iv. 270
```

5.3 Question 3

[3]:
```
friends = {}

n = int(input("Enter number of friends: "))

for i in range(n):
    name = input("Enter friend name: ")
    phone = input("Enter phone number: ")
    friends[name] = phone

print("i. Friends:")
for name in friends:
    print(name, friends[name])

name = input("Enter new friend name: ")
phone = input("Enter phone number: ")
friends[name] = phone
print("ii.", friends)

name = input("Enter friend to delete: ")
if name in friends:
    del friends[name]
    print("Friend deleted")
else:
    print("Friend not found")

name = input("Enter friend to modify: ")
if name in friends:
    friends[name] = input("Enter new phone number: ")
    print("Phone number updated")
else:
    print("Friend not found")

name = input("Enter friend to search: ")
if name in friends:
    print("Friend is present")
else:
    print("Friend is not present")

print("vi. Sorted dictionary:")
for name in sorted(friends):
    print(name, friends[name])
```

```
Enter number of friends:  2
Enter friend name:  raj
Enter phone number:  9876543210
Enter friend name:  suraj
```

```
Enter phone number:  1234567890

i. Friends:
raj 9876543210
suraj 1234567890

Enter new friend name:  sam
Enter phone number:  2345678901

ii. {'raj': '9876543210', 'suraj': '1234567890', 'sam': '2345678901'}

Enter friend to delete:  raj

Friend deleted

Enter friend to modify:  sam
Enter new phone number:  9876543210

Phone number updated

Enter friend to search:  suraj

Friend is present
vi. Sorted dictionary:
sam 9876543210
suraj 1234567890
```

5.4 Question 4

[4]:
```
devices = {
    "D001": {"Device Name": "Smart Light", "Room": "Living Room", "Power␣
 ↪Status": "ON", "Energy Consumption": 12.5},
    "D002": {"Device Name": "Smart TV", "Room": "Living Room", "Power Status":␣
 ↪"OFF", "Energy Consumption": 20.8},
    "D003": {"Device Name": "Smart AC", "Room": "Bedroom", "Power Status":␣
 ↪"ON", "Energy Consumption": 48.3},
    "D004": {"Device Name": "Air Purifier", "Room": "Bedroom", "Power Status":␣
 ↪"ON", "Energy Consumption": 8.5},
    "D005": {"Device Name": "Smart Refrigerator", "Room": "Kitchen", "Power␣
 ↪Status": "ON", "Energy Consumption": 35.7},
    "D006": {"Device Name": "Microwave Oven", "Room": "Kitchen", "Power Status":
 ↪ "OFF", "Energy Consumption": 0},
    "D007": {"Device Name": "Smart Water Heater", "Room": "Bathroom", "Power␣
 ↪Status": "ON", "Energy Consumption": 18.6},
    "D008": {"Device Name": "Smart Door", "Room": "Garage", "Power Status":␣
 ↪"OFF", "Energy Consumption": 5.1},
    "D009": {"Device Name": "Smart Speaker", "Room": "Study Room", "Power␣
 ↪Status": "ON", "Energy Consumption": 3.2},
    "D010": {"Device Name": "Smart Ceiling Fan", "Room": "Dining Room", "Power␣
 ↪Status": "ON", "Energy Consumption": 10.4}
}
```

```
print("i. All devices:")
for i in devices:
    print(i, devices[i])

id = input("ii. Enter Device ID: ")
if id in devices:
    print(devices[id])
else:
    print("Device not found")

print("iii. Devices that are ON:")
for i in devices:
    if devices[i]["Power Status"] == "ON":
        print(i, devices[i]["Device Name"])

total = 0
for i in devices:
    total = total + devices[i]["Energy Consumption"]

print("iv. Total energy:", total)

max_id = ""
max_energy = 0

for i in devices:
    if devices[i]["Energy Consumption"] > max_energy:
        max_energy = devices[i]["Energy Consumption"]
        max_id = i

print("v. Maximum energy device:", max_id, devices[max_id])

id = input("vi. Enter Device ID: ")
if id in devices:
    devices[id]["Power Status"] = input("Enter ON or OFF: ").upper()
    print(devices[id])

id = input("vii. Enter new Device ID: ")
name = input("Enter device name: ")
room = input("Enter room: ")
status = input("Enter power status: ").upper()
energy = float(input("Enter energy consumption: "))

devices[id] = {
    "Device Name": name,
    "Room": room,
    "Power Status": status,
```

```
    "Energy Consumption": energy
}

print("Device added")

id = input("viii. Enter Device ID to delete: ")
if id in devices:
    del devices[id]
    print("Device deleted")
else:
    print("Device not found")

print("ix. Devices in Living Room:")
for i in devices:
    if devices[i]["Room"] == "Living Room":
        print(i, devices[i])

print("x. Sorted dictionary:")
for i in sorted(devices):
    print(i, devices[i])
```

```
i. All devices:
D001 {'Device Name': 'Smart Light', 'Room': 'Living Room', 'Power Status': 'ON',
'Energy Consumption': 12.5}
D002 {'Device Name': 'Smart TV', 'Room': 'Living Room', 'Power Status': 'OFF',
'Energy Consumption': 20.8}
D003 {'Device Name': 'Smart AC', 'Room': 'Bedroom', 'Power Status': 'ON',
'Energy Consumption': 48.3}
D004 {'Device Name': 'Air Purifier', 'Room': 'Bedroom', 'Power Status': 'ON',
'Energy Consumption': 8.5}
D005 {'Device Name': 'Smart Refrigerator', 'Room': 'Kitchen', 'Power Status':
'ON', 'Energy Consumption': 35.7}
D006 {'Device Name': 'Microwave Oven', 'Room': 'Kitchen', 'Power Status': 'OFF',
'Energy Consumption': 0}
D007 {'Device Name': 'Smart Water Heater', 'Room': 'Bathroom', 'Power Status':
'ON', 'Energy Consumption': 18.6}
D008 {'Device Name': 'Smart Door', 'Room': 'Garage', 'Power Status': 'OFF',
'Energy Consumption': 5.1}
D009 {'Device Name': 'Smart Speaker', 'Room': 'Study Room', 'Power Status':
'ON', 'Energy Consumption': 3.2}
D010 {'Device Name': 'Smart Ceiling Fan', 'Room': 'Dining Room', 'Power Status':
'ON', 'Energy Consumption': 10.4}

ii. Enter Device ID:  D004

{'Device Name': 'Air Purifier', 'Room': 'Bedroom', 'Power Status': 'ON', 'Energy
Consumption': 8.5}
iii. Devices that are ON:
D001 Smart Light
```

```
D003 Smart AC
D004 Air Purifier
D005 Smart Refrigerator
D007 Smart Water Heater
D009 Smart Speaker
D010 Smart Ceiling Fan
iv. Total energy: 163.1
v. Maximum energy device: D003 {'Device Name': 'Smart AC', 'Room': 'Bedroom',
'Power Status': 'ON', 'Energy Consumption': 48.3}

vi. Enter Device ID:  D008
Enter ON or OFF:  OFF

{'Device Name': 'Smart Door', 'Room': 'Garage', 'Power Status': 'OFF', 'Energy
Consumption': 5.1}

vii. Enter new Device ID:  D009
Enter device name:  fan
Enter room:  bedroom
Enter power status:  off
Enter energy consumption:  10

Device added

viii. Enter Device ID to delete:  D009

Device deleted
ix. Devices in Living Room:
D001 {'Device Name': 'Smart Light', 'Room': 'Living Room', 'Power Status': 'ON',
'Energy Consumption': 12.5}
D002 {'Device Name': 'Smart TV', 'Room': 'Living Room', 'Power Status': 'OFF',
'Energy Consumption': 20.8}
x. Sorted dictionary:
D001 {'Device Name': 'Smart Light', 'Room': 'Living Room', 'Power Status': 'ON',
'Energy Consumption': 12.5}
D002 {'Device Name': 'Smart TV', 'Room': 'Living Room', 'Power Status': 'OFF',
'Energy Consumption': 20.8}
D003 {'Device Name': 'Smart AC', 'Room': 'Bedroom', 'Power Status': 'ON',
'Energy Consumption': 48.3}
D004 {'Device Name': 'Air Purifier', 'Room': 'Bedroom', 'Power Status': 'ON',
'Energy Consumption': 8.5}
D005 {'Device Name': 'Smart Refrigerator', 'Room': 'Kitchen', 'Power Status':
'ON', 'Energy Consumption': 35.7}
D006 {'Device Name': 'Microwave Oven', 'Room': 'Kitchen', 'Power Status': 'OFF',
'Energy Consumption': 0}
D007 {'Device Name': 'Smart Water Heater', 'Room': 'Bathroom', 'Power Status':
'ON', 'Energy Consumption': 18.6}
D008 {'Device Name': 'Smart Door', 'Room': 'Garage', 'Power Status': 'OFF',
'Energy Consumption': 5.1}
D010 {'Device Name': 'Smart Ceiling Fan', 'Room': 'Dining Room', 'Power Status':
'ON', 'Energy Consumption': 10.4}
```

5.5 Question 5

[5]:
```
food_blogger = {
    "Nagpur": {
        "poha": ["Bardi", "KP Ground"],
        "samosa": ["Shankar Nagar", "Sitabuldi"]
    },
    "Nashik": {
        "misal pav": ["Sadhana", "Shamsundar"],
        "samosa": ["ABC", "XYZ"]
    },
    "Pune": {
        "misal pav": ["Katakirr", "Bedekar"],
        "vada pav": ["Garden Vada Pav", "JJ Garden"]
    },
    "Mumbai": {
        "vada pav": ["Ashok Vada Pav", "Anand Stall"],
        "pav bhaji": ["Sardar Refreshments", "Cannon"]
    },
    "Indore": {
        "poha": ["Joshi Dahi Bada", "Young Tarang"],
        "bhutte ka kees": ["Sarafa Bazaar", "Vijay Chaat House"]
    }
}

city = input("i. Enter city name: ")

if city in food_blogger:
    print(food_blogger[city])
else:
    print("City not found")

food = input("ii. Enter food specialty: ")

for city in food_blogger:
    if food in food_blogger[city]:
        print(city)

city = input("iii. Enter city name: ")
food = input("Enter food specialty: ")

if city in food_blogger and food in food_blogger[city]:
    print(food_blogger[city][food])
else:
    print("Not found")
```

```
i. Enter city name:  pune
City not found
```

```
ii. Enter food specialty:  vada pav

Pune
Mumbai

iii. Enter city name:  Pune
Enter food specialty:  poha

Not found
```

5.6 Practice Question 1

[6]:
```
C =␣
  ↪{203,204,207,216,217,219,226,227,230,233,239,248,250,254,260,262,272,279,281,282,289,291,29

F =␣
  ↪{200,202,203,220,222,225,226,229,237,241,252,253,260,262,264,269,270,276,282,283,289,290,29

R =␣
  ↪{203,204,205,207,211,212,213,220,223,225,226,229,233,235,244,245,246,247,248,251,252,254,25

print("i.", len(C & F & R))

only_C = C - F - R
only_F = F - C - R
only_R = R - C - F

print("ii.", len(only_C | only_F | only_R))
print("iii.", len((C & F) | (C & R) | (F & R)))
```

```
i. 9
ii. 134
iii. 58
```

5.7 Practice Question 2

[7]:
```
Employee_data = {
    101: ['Shiva', 24, 'Content Strategist'],
    102: ['Udit', 25, 'Content Strategist'],
    103: ['Sonam', 28, 'Sr Manager'],
    104: ['Ansari', 29, 'Product Lead'],
    105: ['Huzefa', 32, 'Project Manager']
}

oldest = max(Employee_data, key=lambda x: Employee_data[x][1])
print("i.", Employee_data[oldest])

if 159 in Employee_data:
    print("ii.", Employee_data[159][1])
```

```
else:
    print("ii. NA")

print("iii.", len(Employee_data))

total = 0
for i in Employee_data:
    total = total + Employee_data[i][1]

print("iv.", total / len(Employee_data))

for i in [104, 140, 164]:
    if i in Employee_data:
        Employee_data[i][1] = 27

total = 0
for i in Employee_data:
    total = total + Employee_data[i][1]

print("v.", total / len(Employee_data))
```

```
i. ['Huzefa', 32, 'Project Manager']
ii. NA
iii. 5
iv. 27.6
v. 27.2
```

5.8 Practice Question 3

[8]:
```
input_dict = {
    'Jack Dorsey': 'Twitter',
    'Tim Cook': 'Apple',
    'Jeff Bezos': 'Amazon',
    'Mukesh Ambani': 'RJIO'
}

print(sorted(input_dict.values()))
```

```
['Amazon', 'Apple', 'RJIO', 'Twitter']
```

5.9 Practice Question 4

[9]:
```
Olympic = {
    "Great Britain": {"GBR": {"Gold": 29, "Silver": 17, "Bronze": 19}},
    "China": {"CHN": {"Gold": 38, "Silver": 28, "Bronze": 22}},
    "Russia": {"RUS": {"Gold": 24, "Silver": 25, "Bronze": 32}},
    "United States": {"USA": {"Gold": 46, "Silver": 28, "Bronze": 29}},
    "Korea": {"KOR": {"Gold": 13, "Silver": 8, "Bronze": 7}},
```

```
    "Japan": {"JPN": {"Gold": 7, "Silver": 14, "Bronze": 17}},
    "Germany": {"GER": {"Gold": 11, "Silver": 11, "Bronze": 14}}
}

max_gold = 0
max_country = ""

for country in Olympic:
    code = list(Olympic[country])[0]
    gold = Olympic[country][code]["Gold"]
    if gold > max_gold:
        max_gold = gold
        max_country = country

print("i.", max_country)

print("ii.")
for country in Olympic:
    code = list(Olympic[country])[0]
    if Olympic[country][code]["Gold"] > 20:
        print(country)

print("iii.")
for country in Olympic:
    code = list(Olympic[country])[0]
    gold = Olympic[country][code]["Gold"]
    silver = Olympic[country][code]["Silver"]
    bronze = Olympic[country][code]["Bronze"]
    total = gold + silver + bronze
    print(country, gold, total)
```

```
i. United States
ii.
Great Britain
China
Russia
United States
iii.
Great Britain 29 65
China 38 88
Russia 24 81
United States 46 103
Korea 13 28
Japan 7 38
Germany 11 36
```

5.10 Practice Question 5

[14]:
```python
def mail_id(name):
    a = name.lower().split()
    return a[2] + a[0][0] + a[1][0] + "@rknec.edu"

students = {}

for i in range(5):
    name = input("Enter full name: ")
    students[mail_id(name)] = name

print(students)
```

```
Enter full name:  Meet Mukesh Agrawal
{'agrawalmm@rknec.edu': 'Meet Mukesh Agrawal'}
```

[ ]: